BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: PR–X (A-I-B6) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR–X (A-I-B6) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. N2 is converted to ammonium with the help of a bacteria belonging to the genus;

(A) Pseudomonas
(B) Rhizobium
(C) Nitrosomonas
(D) Nitrobacter

Q2. Which of the given cause stunted growth in plants?

(A) Deforestation
(B) Acid rain
(C) Global warming
(D) Urbanization

Q3. The technique used to make the plants resistant to insects is;

(A) Artificial selection
(B) Carbohydrates transfer
(C) Fats transfer
(D) DNA transfer

Q4. Narcotics most generally come from;

(A) Cannabis sativa
(B) Mushroom
(C) Opium
(D) Cactus

Q5. The insertion of bicep muscle is;

(A) Ulna
(B) Radius
(C) Femur
(D) Humor

Q6. Development of new individual from female gamete without fertilization is;

(A) Budding
(B) Germination
(C) Parthenogenesis
(D) Pollination

Q7. Based on 2017 census, Pakistan's population is about 208 million where the growth rate of Khyber Pakhtunkhwa is estimated to be;

(A) 2.49%
(B) 2.69%
(C) 2.89%
(D) 2.99%

Q8. Genotype express:

(A) Combination of the alleles of a gene pair
(B) Physical appearance of the alleles of a gene pair
(C) Physical appearance of a gene
(D) Physical combination of dominant allele

Q9. A sheet of skeletal muscles forming the floor of chest cavity is

(A) Rib
(B) Sternum
(C) Trachea
(D) Diaphragm

Q10. The non-surgical technique that helps to remove kidney stones is:

(A) Urination
(B) Dialysis
(C) Lithotripsy
(D) Transplantation

Q11. The largest part of brain is:

(A) Cerebrum
(B) Cerebellum
(C) Thalamus
(D) Pons

Q12. Myelin sheath is made of:

(A) Carbohydrates materials
(B) Protein materials
(C) Fatty materials
(D) Phosphate materials

SECTION-B

Time Allowed: 2 Hours 45 Minutes

> Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 45 minutes.

Q2. Answer any EIGHT parts. Each part carries FOUR marks.

i) Write FOUR main causes of lungs cancer.

ii) How Halophytes reduce loss of water from their bodies and absorb salt from surrounding environment?

iii) Compare the function of glomerulus and rental tubule.

iv) Differentiate between electrical coordination and hormonal coordination with special reference to their receptors.

v) What happens if thyroxin is produced more than the required amount?

vi) Briefly explain the factors which determine the flexibility of a cartilage?

vii) Briefly discuss the future of the following structure after fertilization. (a) Zygote (b) Triploid Nucleus

viii) Highlight the role of mRNA and tRNA in protein synthesis.

ix) Briefly explain water as an important abiotic factor.

x) Differentiate between alcoholic fermentation and lactic acid fermentation in terms of start and end products.

xi) Briefly explain narcotics as addictive drug in terms of: (a) Acting on body parts (b) Effects on body

SECTION-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 16

> Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) Briefly explain the processes which are involved in the excretion of extra water from plant bodies.

(b) List any THREE emergency aid to a patient who suffered from a seizure.

Q4.

(a) Differentiate between the joint of vertebral column and Hip joint.

(b) Write name of any three components of male reproductive system of rabbit.

Q5.

(a) How does chromatin material play an important role in the composition of chromosome?

(b) Discuss the role of air and water pollutants in causing land pollution.

Q6.

(a) Discuss the objectives of genetic engineering in terms of Genetically Modified Plants.

(b) Write behavioral symptoms of drug addiction with the help of examples.